WEEK 04 REPORT

EDA DEEP DIVE

MONEYBALL – IPL PLAYER VALUATION

Data Science and Visualization | Course Code: 24ADI204

Data-Viz Hack-A-Bit (2026–2027)

TEAM NUMBER: 05

Team Member	Register Number
Divyesh D S	25BAD022
Kavi Prakash M	25BAD043
Mithuin P S	25BAD057
Madhan P M	25BAD503
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Univariate and Bivariate Analysis – Visualizing Distributions

1. Introduction

Moneyball – IPL Player Valuation explores Indian Premier League player performance records alongside available player-profile, team and auction-related information. The project aims to understand how performance indicators relate to player valuation and to develop insights for a later dashboard.

Week 04 focuses on Exploratory Data Analysis (EDA), following the Week 03 cleaning activity. It examines individual variable distributions and relationships between pairs of variables using statistical summaries and visualizations.

2. Problem Statement

The IPL dataset contains batting, bowling, player-profile and auction-related attributes. Individual records alone do not reveal the overall distribution of these measures or the relationships between them. EDA helps identify patterns, variability and associations that can guide later feature engineering and player-valuation analysis.

3. Objectives

• Verify the cleaned dataset structure.
• Separate numerical and categorical variables.
• Examine distributions using histograms, KDE plots, boxplots and count plots.
• Explore relationships using scatter plots and correlation analysis.
• Study season-wise performance and leading run scorers.
• Record findings without changing the cleaned dataset.

4. Team Formation

Team Member	Register Number
Divyesh D S	25BAD022
Kavi Prakash M	25BAD043
Mithuin P S	25BAD057
Madhan P M	25BAD503
Renganathan G M	25BAD087

Specific individual task allocations were not provided in the reviewed materials, so individual responsibilities are not assigned.

5. Dataset Profile

The Week 04 activity uses **IPL_Moneyball_Cleaned_Week03.csv**, carried forward from Week 03. The dataset contains player-season records and batting, bowling, team, player-profile and auction-related attributes.

Property	Observed value
Rows	3,311
Columns	27
Numerical columns	17
Categorical columns	10
Season coverage	2008–2026 (19 seasons)
Distinct player values	794
Missing cells	13,099
Duplicate rows	0

The 27 columns include season, player, runs, balls_faced, innings, fours, sixes, dismissals, balls_bowled, runs_conceded, wickets, matches_bowled, auction_name, team, role, nationality, status, acquisition_type, base_price_lakh, final_price_lakh, bat_style, bowl_style, player_full_name, strike_rate, batting_avg, economy and bowling_avg.

6. Tools and Libraries

Python: execution environment. **Pandas:** loading, grouping, summaries and correlation calculations. **NumPy:** numerical operations. **Matplotlib:** chart creation. **Seaborn:** statistical visualizations. The workflow was carried out in Google Colab.

7. Methodology and Implementation

7.1 Dataset loading and verification

The cleaned Week 03 CSV was loaded into a Pandas DataFrame. The notebook checked its dimensions, columns, data types and records to confirm that the same project dataset was used.

7.2 Numerical and categorical analysis

Numerical and categorical columns were identified separately. Descriptive statistics were used to inspect count, mean, standard deviation, minimum, quartiles and maximum. Skewness and kurtosis were calculated to describe distribution asymmetry and tail behaviour.

7.3 Univariate analysis

Histograms and KDE plots were prepared for runs, strike rate, wickets and economy. Boxplots were used for runs, strike rate, batting average, wickets, economy and final price. A role count plot was used to compare category frequencies.

7.4 Bivariate analysis

Scatter plots examined runs against balls faced, runs against fours, runs against sixes, wickets against runs conceded, and batting average against strike rate. These plots help identify the direction and strength of pairwise patterns.

7.5 Correlation and season analysis

Pearson correlation coefficients were calculated for numerical variables and visualized with a heatmap. Season-wise totals were grouped to examine performance across seasons. Player records were grouped to identify leading aggregate run scorers.

8. Results and Analysis

8.1 Data-quality verification

Measure	Week 03	Week 04
Rows	3,311	3,311
Columns	27	27
Missing cells	13,099	13,099
Duplicate rows	0	0

The values remained unchanged because Week 04 was an analysis-only activity. No additional cleaning, imputation, deletion or transformation was applied.

8.2 Correlation results

Variable pair	Correlation
Runs vs Balls Faced	0.983936
Balls Bowled vs Runs Conceded	0.982674
Balls Bowled vs Matches Bowled	0.976987
Runs vs Fours	0.964513

These are very strong positive linear associations. For example, runs increase with balls faced in the dataset, while runs and fours also move together. Correlation indicates association; it does not prove that one variable causes the other.

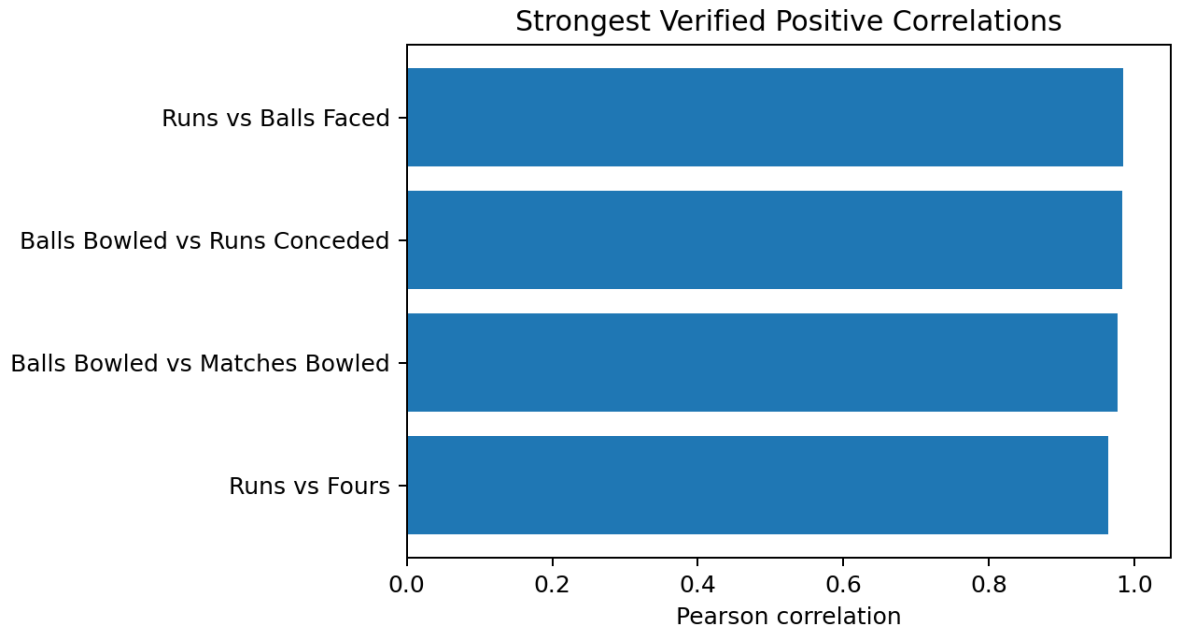


Figure 1. Strongest positive correlations reported by the Week 04 analysis.

8.3 Top players by aggregate runs

Rank	Player	Total runs
1	V Kohli	9,040
2	RG Sharma	7,183
3	S Dhawan	6,769

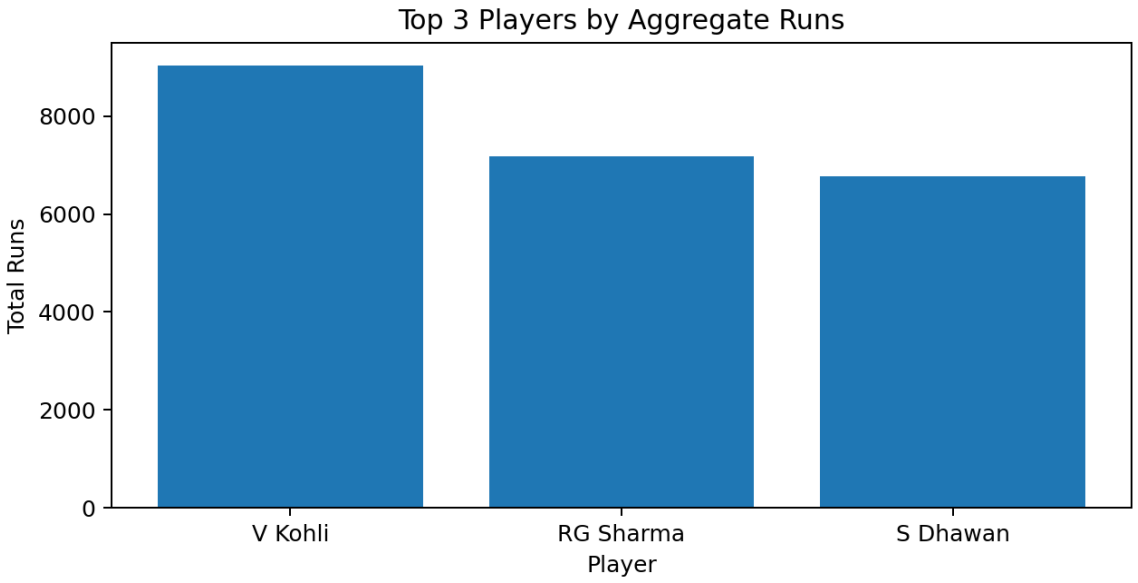


Figure 2. Top three players by aggregate runs; only verified values are plotted.

9. Visual Evidence and Interpretation

The figures above visualize verified numeric outputs. The Week 04 notebook also contains histograms, KDE curves, boxplots, a role count plot, scatter plots, a correlation heatmap and a season-wise line plot. These notebook plots should be retained as the detailed execution evidence.

Histograms and KDE curves describe distribution shape. Boxplots show medians, interquartile ranges and potential extreme observations. Scatter plots show pairwise patterns, while the heatmap makes numerical associations easier to compare. Extreme values should be investigated in context rather than automatically removed, because exceptional IPL performances may be genuine.

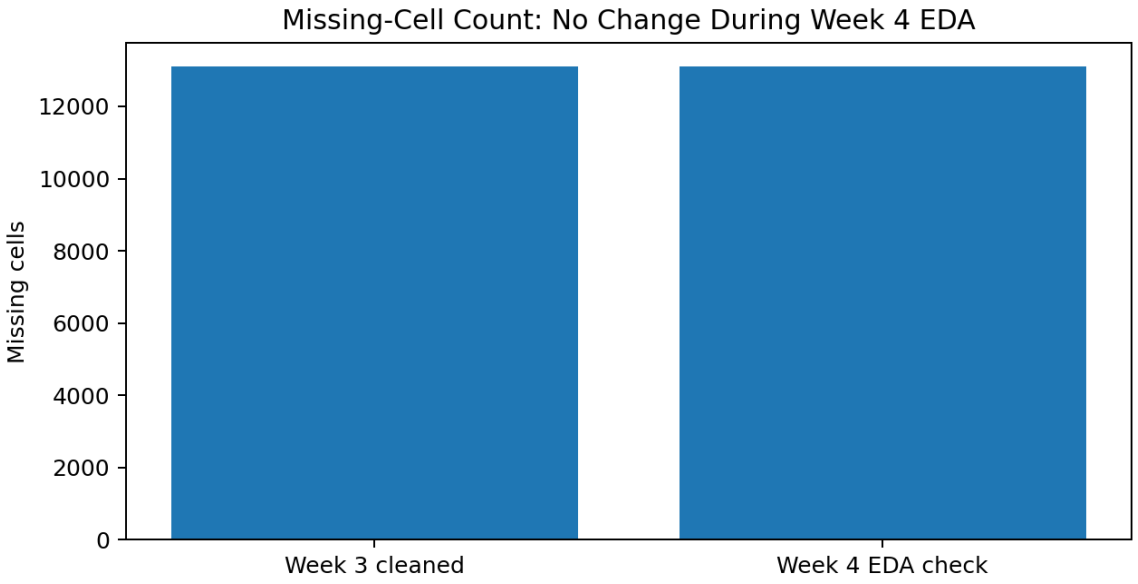


Figure 3. Missing-cell count remained the same between Week 03 and Week 04.

10. Before-and-After Comparison

Week 04 did not change the Week 03 cleaned dataset. The unchanged row count, column count, missing-cell count and duplicate count are expected for an EDA-only stage.

11. Findings and Discussion

• The dataset contains 3,311 player-season records across 19 seasons.
• Runs and balls faced have a very strong positive association (0.983936).
• Balls bowled and runs conceded have a very strong positive association (0.982674).
• Runs and fours have a very strong positive association (0.964513).
• Distribution plots provide information about spread and skewness beyond averages.
• Remaining missing values must be considered carefully in later modelling.
• Aggregate runs alone do not determine player value.

12. Limitations

The analysis is limited to the available fields and records in the team-provided merged dataset. Correlation does not establish causation. Aggregate run totals do not capture every factor relevant to valuation, including player role, match context, strike rate, bowling contribution, availability and auction conditions. The report therefore presents descriptive findings, not a completed valuation model.

13. Conclusion

Week 04 completed EDA through univariate and bivariate analysis, distribution visualization, correlation analysis and season-wise exploration. The cleaned dataset contained 3,311 rows and 27 columns. Verification showed 13,099 missing cells and zero duplicate rows, unchanged during this analysis-only stage. The observed relationships provide a foundation for later feature engineering and dashboard development.

14. Project Files and GitHub

The Week 04 GitHub folder should contain exactly two files:
`Week_04/Week04_EDA_Deep_Dive.ipynb`
`Week_04/Week04_Report.pdf`

Repository: https://github.com/Mithuinps/24ADI204-DSV-team-05/tree/main

15. References

1. Week 04 EDA notebook: Week04_EDA_Deep_Dive.ipynb.
2. Dataset: IPL_Moneyball_Cleaned_Week03.csv.
3. Data-Viz Hack-A-Bit (2026–2027), Data Science and Visualization, course code 24ADI204, Week 04 activity.

4. Project GitHub repository: https://github.com/Mithuinps/24ADI204-DSV-team-05/tree/main.

Note: Graphs in this PDF are generated from the verified numeric results available for this report. Raw row-level dataset data was not available here to reconstruct additional histograms or scatter-point plots. The executed notebook remains the source for those full plots.